Jiaxun Li

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Statistics PhD student Statistical learning theory, generative modeling, and ML evaluation.
Seeking Summer 2027 research internships in machine learning and AI foundations.

Education

University of Michigan, Ann Arbor Sep. 2024 – expected Sep. 2029
Ph.D. in Statistics Advisors: Ambuj Tewari and Saptarshi Chakraborty

Xi'an Jiaotong University Sep. 2020 – Aug. 2024
B.Sc. in Mathematics, Honors Program GPA: 4.21/4.30

Selected Research and Publications

Generation through the lens of learning theory *COLT 2025*
Jiaxun Li*, Vinod Raman*, Ambuj Tewari *Equal contribution

- Developed mathematical frameworks and theoretical characterizations of generation, connecting generative capabilities with statistical learning theory and online learnability.
- Presented the work as a contributed presentation at COLT 2025.

On generation in metric spaces *NeurIPS 2026, forthcoming*
Jiaxun Li, Vinod Raman, Ambuj Tewari

- Studied generation in metric spaces through complexity measures and mathematical proofs, characterizing how geometry and novelty scales affect generatability.

Robust Detection of LLM-Generated Text under Contamination *Under review, 2026*
Jiaxun Li, Saptarshi Chakraborty, Ambuj Tewari

- Independently designed the benchmark and implemented models, experimental pipelines, and Python evaluations of LLM-text detection under contamination.
- Evaluated detector performance using AUROC and detection at low false-positive rates; personally directed the experiments and analysis.

Exact and conservative inference for the average treatment effect in stratified experiments with binary outcomes *Under review, 2025*
Jiaxun Li, Jacob Spertus, Philip B. Stark

- Implemented simulation experiments for research on statistical inference in stratified randomized experiments with binary outcomes.

Technical Skills

Programming: Python, R, MATLAB, C/C++
Libraries: NumPy, SciPy, PyTorch, scikit-learn **Tools:** Bash, Git, LaTeX

Teaching and Recognition

Graduate Student Instructor, University of Michigan 2024 – 2026
Probability and Distribution Theory; Capstone Seminar; undergraduate probability and statistics.
Outstanding Graduate Student Instructor Team Award, STATS 485 (Winter 2026).